# ZAMBIA UNIVERSITY COLLEGE OF TECHNOLOGY

## ICT 2205 Data Structures and Algorithms

### Final examination - Academic year 2022, semester 1

Time allowed: 3 hours  Total marks: 100  Lecturer: Mrs J. Chibwe

*Instructions: answer ALL questions. Write clearly. Show all working.*

### Question 1: Linked lists

(a) Inserting at the head of a singly linked list takes:

(b) Each node of a doubly linked list stores:

(c) Finding the k-th element of a linked list takes:

(d) A circular linked list's last node points to:

### Question 2: Sorting

(a) Which sort has worst case O(n^2) but average O(n log n)?

(b) Which sorting algorithm is stable and always O(n log n)?

(c) Bubble sort's best case on an already sorted list (with early exit) is:

(d) Which sort repeatedly selects the minimum and places it at the front?

### Question 3: Trees

(a) In-order traversal of a binary search tree visits keys in:

(b) The height of a balanced binary tree with n nodes is about:

(c) A binary heap is usually stored in:

(d) Which tree keeps itself balanced using rotations based on height differences?

*END OF PAPER - generated sample for the ZutLib prototype*